Additive Manufacturing

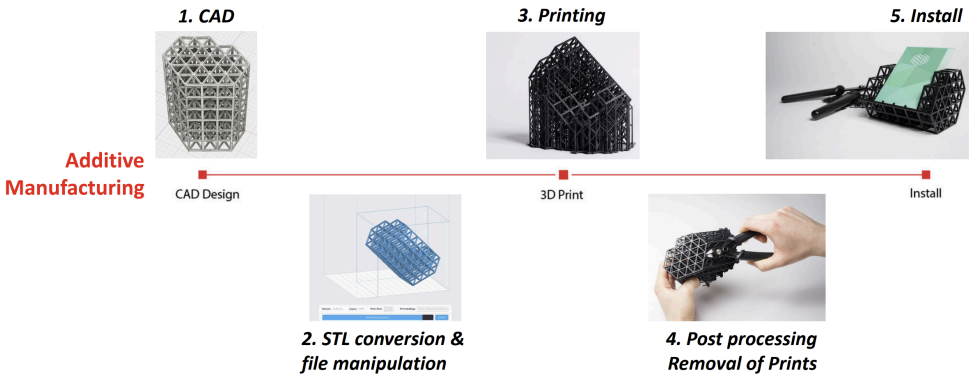
Additive manufacturing, or 3D Printing is a method which layers are built on top of one another to form solid objects, useful for rapid prototyping or even functional parts.

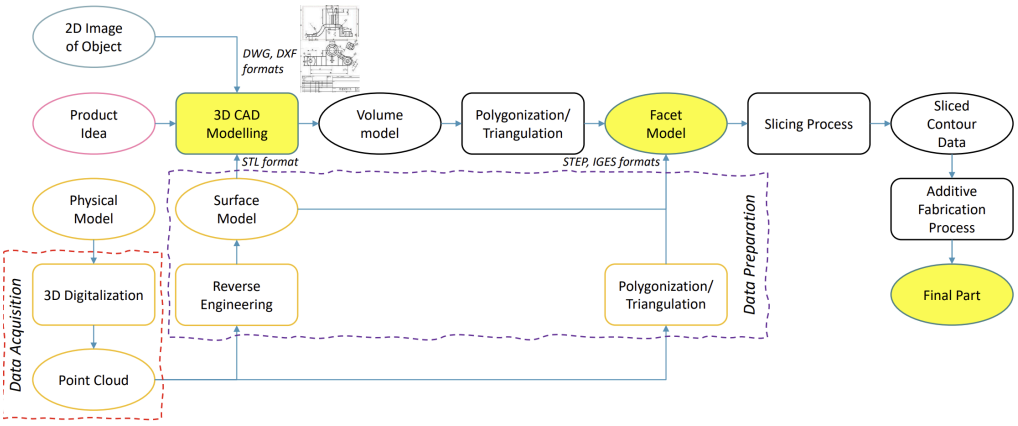
AM has a lower cost for prototyping, start ups, can make complex forms, and works with a wide range of materials, however has lower strength while also being anisotropic (directional strength), and has a lower accuracy (Post processing is also required to remove supports or to achieve a certain finish, and the cost benefit becomes less significant at larger volumes.

Despite the disadvantages, AM is still used in automotive, healthcare, product design and robotics applications.

Additive Manufacturing Process

The typical AM process goes like so:





CAD Modelling

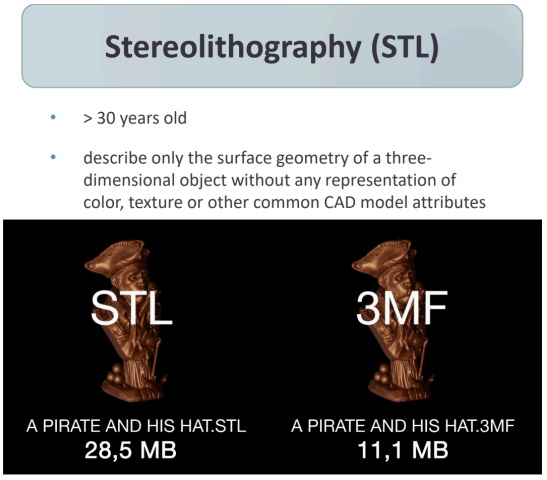
A 3D CAD model is first made, either with CAD software or reverse engineering methods, like 3D scanning

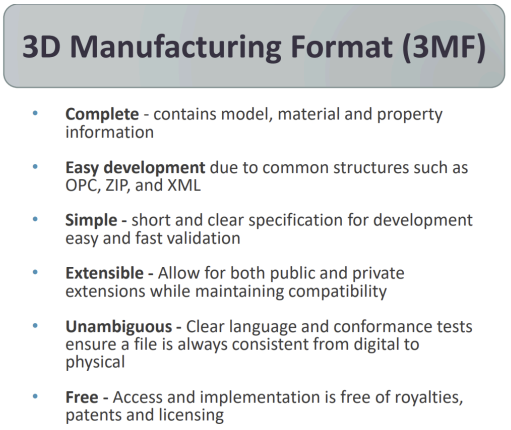
CAD to STL/3MF

The cad model is converted to a .stl file via tesselation, which converts the model into a series of triangles, stored as the three vertices' coordinates and the surface normal vector. however, sometimes the triangle data may have defects

triangle size depends on the chordal tolerance set, smaller tolerance creates smaller triangles which captures more details but causes larger file size.

recently, newer alternative to stls is 3MF, which contains more information about the print, while hvaing a smaller file size:





Slicing

slicing is the process of converting the stl file into gcode. The working principle is that the contour at each slice of the model is captured.

the thickness of each slice can be controlled, where thicker slices cause faster print time but poorer print quality and resolution

overhangs are also mitigated in this step with support structures.

overhangs are unfeasible and sometimes impossible to print because of the lack of material below it, which leads to curling, sagging, delamination and collapsing

to mitigate this, support structures that are removed after can be added, at the expense of poor surface finish and more print time. overhangs can also be reduces to less than 45 degrees.

Printing

occurs from the bottom up in layers (no shit)

Post Processing

any processes that need to be done afterwards before use, like removing supports, filling voids, sanding, or adding paint

As there are multiple types of additive manufacturing, we can use a similar PRIMA matrix to pick the best method depending on the material required and the speed of production needed

MATERIAL / SPEED	PAPER	SAND	PLASTER	CELLULOSE	STARCH	WAX	ELAST-OMER	EPOXY	ABS	PA	PC	PE	PPS	PS	PU	PVC	UP	IRON - COPPER ALLOYS	STEEL (carbon)	STEEL (tool, alloy)	STAINLESS STEEL	COPPER ALLOYS	ALUMINIUM	NICKEL ALLOYS	TITANIUM ALLOYS	TUNGSTEN	CERAMICS	METAL-CERAMIC COMPOSITES
SLOW Volume 0.05 ℓ/h Vertical Build 1.5mm/h						[8.5]	[8.5]		[8.5]	[8.5]	[8.5]	[8.5]	[8.5]				[8.5]										[8.5]	
MEDIUM Volume 0.1 ℓ/h Vertical Build 5mm/h	[8.4]	[8.3]				[8.3]	[8.1] [8.3]	[8.1]	[8.4]	[8.3]	[8.3] [8.4]			[8.3] [8.4]	[8.1]	[8.3] [8.4]		[8.3]	[8.3] [8.4]	[8.3]	[8.3]	[8.3]	[8.3] [8.4]	[8.3]	[8.3]	[8.3]	[8.4]	
FAST Volume 0.25 ℓ/h Vertical Build 20mm/h		[8.2]	[8.2]	[8.2]	[8.2]	[8.2]	[8.2]	[8.2]							[8.2]						[8.2]	[8.2]					[8.2]	[8.2]

KEY TO RAPID PROTOTYPING PROCESS PRIMA SELECTION MATRIX:

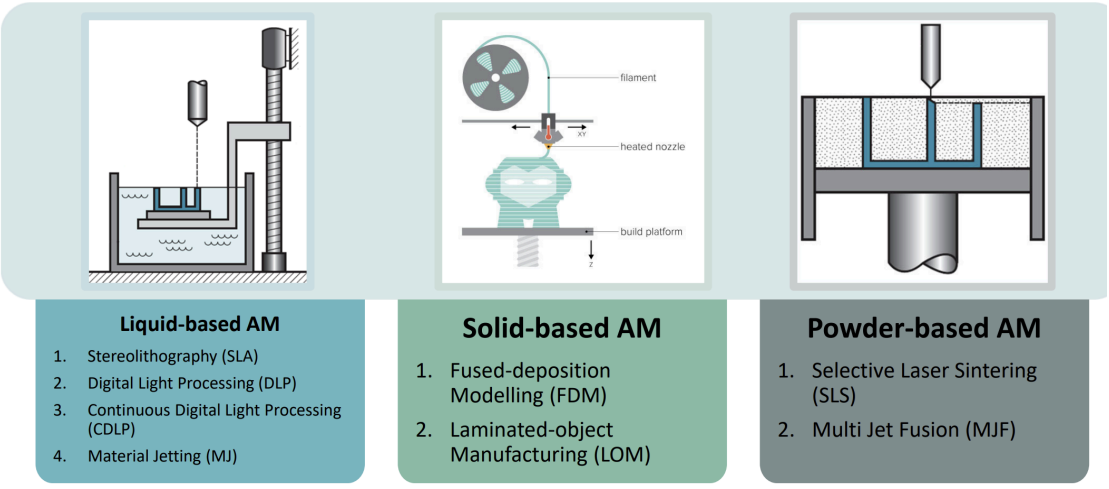
RAPID PROTOTYPING PROCESSES

[8.1] STEREOLITHOGRAPHY (SLA)
[8.2] MULTI-JET FUSION (MJF)
[8.3] SELECTIVE LASER SINTERING (SLS)
[8.4] LAMINATED OBJECT MANUFACTURING (LOM)
[8.5] FUSED DEPOSITION MODELLING (FDM)

MATERIALS

ABS Acrylonitrile Butadiene Styrene
PA Polyamide
PC Polycarbonate
PE Polyethylene
PPS Polyphenylenesulphone
PS Polystyrene
PU Polyurethane
PVC Polyvinylchloride
UP Polyester

Shown are some types of AM:



But there are only a few methods that are the most common:

FDM (Fused Deposition Modelling)

the most common type of AM, where a thermoplastic is fed into a heated extruder which deposits the plastic onto the print bed to form layers. Typically, a 0.4mm nozzle is used with filament of 1.75mm or 3mm diameter

Filament Materials	Thermoplastics (PLA, ABS, PETG, PC, PEI etc)
Filament Diameters	• 1.75 mm • 3.00 mm
Dimensional accuracy	• ± 0.5% (lower limit ± 0.5 mm) – desktop • ± 0.15% (lower limit ± 0.2 mm) - industrial
Typical build size	• 200 x 200 x 200 mm – desktop • 1000 x 1000 x 1000 mm - industrial
Common layer height	50 to 400 microns
Support	Not always required (dissolvable available)

FDM works with a wide range of thermoplastics, from common plastics like PLA and ABS, to engineering thermoplastics with special properties like TPU and PETG:

	Material	Characteristics
	PLA	✓ Excellent visual quality ✓ Easy to print × Low impact strength
	ABS	✓ Good strength. ✓ Good temperature resistance. × More susceptible to warping.
	PETG	✓ Food Safe* ✓ Good strength ✓ Easy to print with
	Nylon (PA)	✓ High strength ✓ Excellent wear and chemical resistance × Low humidity resistance
	TPU	✓ Very flexible × Difficult to print accurately
	PEI	✓ Excellent strength to weight ✓ Excellent fire and chemical resistance × High cost

Some parameters and effects of FDM are as shown:

Infill & Shell Thickness

higher infill and shell thickness causes higher strenght but more print time

Warping

Uneven cooling of the part causes internal stresses which causes deformations, like corners being lifted from the build plate. This problem is more prominent on ABS and PC (polycarbonate), compared to PLA

Layer Adhesion

Layer adhere to each other due to the high temperature of new layers remelting the previous layer together, enabling bonding.

If the new layer is insufficiently pressed against the previous layer or the build plate, poor layer adhesion occurs which prevents bonding from taking place, causing layer shifts and spaghettification

Overhangs

overhangs are caused by insufficient support from the layer below, requiring supports of overhangs more than 45 degrees

SLA & DLP (Stereolithography & Digital Light Processing)

Both methods work by curing liquid resin onto a build plate layer by layer, with either a single point UV light or a digital light projector, respective to SLA and DLP

Typically, parts are printed at angles to reduce stresses on the part.

Afterwards, the part is cleaned and cured with an UV light, then supports are removed and post processing is done for high surface finish

SLA is able to achieve high accuracy, intricate details, smooth surface, with a large range of specialty materials. However, parts are brittle, degrades when exposed to sunlight, and support marks need to be removed.

MJ (Material Jetting)

MJ works with multiple print heads jetting hundreds of tiny droplets of phot polymer on the build plate to form a layer , and the part is cured with UV light, before being washed with water to dissolve the water soluble supports.

High accuracy, fine details, injection mould like finish, multi materials and full colors can be achieved with this method, at the expense of cost, mechanical properties degrading over time, and brittle parts.

SLS (Selective Laser Sintering)

A powder bin is heated to below its melting point and a blade/roller deposits a thin layer of powder over the bed, before a CO2 laser scans the powder bed and sinters the particles selectively and binds them together to form a layer.

After the part is formed, the powder is cooled and the non sintered powder is removed and cleaned out (to be reused)

This method is suitable for nylon, TPU, carbon filled, glass filled and PA11.

This method is ideal for functional prototypes with good mechanical properties, complex geometries w/o support, and for small batch production, however has higher cost, slower production time, and a grainy surface with internal porosity due to the powder and the sintering process.

To design parts for AM, we have to take note of a few design guidelines, depending on the AM process used.

	FDM	SLA	SLS	MJ
Min Supported Walls	0.8	0.5	0.7	1
Min Unsupported Walls	0.8	1	-	1
Max Overhangs	45	always needed	-	always needed
Embosses & Engravings	0.6 wide 2 mm high	0.4 wide 0.4 high	1 wide 1 high	0.5 wide 0.5 high
Max Horizontal Bridges	10mm	-	-	-
Min Holes	2	0.5	1.5	0.5
Min Moving Components Tol.	0.5	0.5	0.1 press fit, 0.3 moving	0.2
Escape Holes (for excess resin or powder to be drained)	-	4	5	-
Minimum Features	2	0.2	0.8	0.5
Min Pin Diameter	3	0.5	0.8	0.5

Tolerances (symmetric, +-)	0.5	0.15	0.3	0.1

Other guidelines involve:

Splitting Model, to allow easier/higher quality printing to then be assembled

Orientating Parts, to remove need for support for holes (typically prioritise blind holes, followed by smaller diameters, or the criticality of the hole)

Build direction, to have layers perpendicular to the direction of load, to take advantage of FDM's anisotropic properties

Optimising Topology, optimising material layout to best support the applied loads, optimising weight reduction, strenght, cost, and performance depending on the initial constraints placed.